# HC Portal 357

UNRESOLVED | Reviewed attempt | 11 October 2026

No cipher key or supported plaintext was recovered. This report closes a reviewed attempt, not a decipherment. There is no register entry, original-language plaintext, or plaintext translation.

## Origin and evidence

The portal identifies this as MZV message 80132/1944, dated 15 March 1944, from Belnay to the Slovak legation in Sofia. Archive: Slovak National Archive in Bratislava, MZV, box 42. The title concerns instructions for handing over decorations; it is contextual metadata, not a decrypted message. The scan includes two 15. III. 1944 stamps and a handwritten receipt time of 17.45.

The source is the portal-displayed 1024-pixel-wide large image conversion. An original-resolution upload was not obtained. No transcription or solution attachment was shown. The image bytes remain unchanged. OpenAI GPT-6 Astra manually transcribed the body; Claude Opus 5.5 independently agreed with all 85 letters. Possible n/m ambiguities remain in groups 5 and 10.

```
amaon yoato qcbzu mtrds ddnmn tooip
yzlro vpatv pnyly hndei raaza oznrc
cyiao ysoce rescm nnaaa vzvvk
```

## Method and result

The header 685 has a suffix matching the 85-letter body. The meaning of the leading 6 remains unresolved. The index of coincidence is 0.05742; removing group 3 yields 0.06234. These statistics are compatible with transposition, but do not identify a cipher.

The historical R/XQ system is a candidate: three columnar transpositions with daily key permutations and a five-letter indicator. Neither its use here nor an indicator position is established. Searches covered simple routes, Caesar/reversal controls, and stochastic single-columnar variants of widths 2-30. Inputs included the full body, removal of group 3, and, after review, removal of group 6. None produced coherent plaintext.

Two- and three-stage searches failed known-answer controls. Their target failures therefore have no evidential weight against multiple transposition or solvability. Single-columnar controls succeeded at width 11; that demonstrates limited capability, not exhaustive coverage of every width and transcription.

## Independent review and corrections

Claude Opus 5.5 returned REVISE in review/REVIEW1.md. The reviewer confirmed the transcription, checksums and byte-identical reproduction of eight output files. The reviewer explicitly required the outcome to remain unresolved and allowed an unresolved report after the corrections; no second round was required unless a plaintext emerged.

1. The report and working notes now clearly separate limited single-columnar evidence from failed multi-stage optimization. Reviewer controls recovered 3/3 single-columnar examples, but 0/3 double and 0/3 triple examples.

2. The suffix/length pattern is recorded. The requested group-6-removal search was completed in both directions. Best scores were 136.605 and 125.829; candidates remain incoherent. These scores are heuristic rankings, not decipherment confidence.

3. Group 3 is no longer supported by the presence of Q: documented preparation permits Q as a plaintext separator. The unique B/U observation is weak and establishes no indicator.

4. Equal stage widths were an unsupported simplification. Figure 5 of the historical paper uses widths 21, 19 and 20. Its example has alignment n=7, indicator position ip=5, and header 360. Consequently a universal rule equating the header prefix with alignment or indicator position is not supported. A different local convention remains possible.

5. The large-conversion source limitation is explicit, both n/m uncertainties and receipt annotations are recorded in a working supplement, and the displayed source hash was corrected without changing source bytes. The full item-by-item response is in SOLUTION.md, section Response to review 1.

## Reproduction and remaining uncertainty

```
sh ciphers/hcp-357/work/reproduce.sh
```

Run from the repository root. This regenerates diagnostic candidates from the manual transcription; it does not decrypt the source into a full plaintext. The exact cipher, daily key, indicator position, original-resolution readings, plaintext and translation are still unknown. Useful progress needs better source/key evidence or an attack validated on comparable multi-stage controls. No mathematical impossibility or unsolvability is claimed.

## Sources and provenance

Portal record: https://crypto.hcportal.eu/dashboard/cryptograms/357
Historical method: Antal, Zajac and Grošek, Diplomatic Ciphers Used by Slovak Attaché During the WW2, HistoCrypt 2020, pp. 22-23 and Figure 5. https://ep.liu.se/ecp/171/004/ecp2020_171_004.pdf

Language scoring used fourgram log counts from Universal Dependencies Slovak SNK training text, documented in SOURCES.md. No published solution of this target was used. Local evidence: source/cryptogram.jpg, NOTES.md, SOLUTION.md, review/REVIEW1.md and work/.